Dear Editor,

Thank you for the opportunity to submit a second revision of our manuscript **ASOC-D-25-13526R2**, entitled "Optimization of Empty Container Repositioning by Integrating Large Language Models and Multi-Agent Systems." We sincerely thank all three reviewers for their insightful comments and valuable suggestions, which have substantially strengthened our manuscript. The major revisions include:

- **Notation and presentation (Reviewer 3):** Unified the agent naming convention (full names with symbolic notation, e.g., "Project Manager Agent (a_{PM})"); defined variable x_{ij}; and formalized the Structural-Agreement Rate (SAR) with all tuple components defined and a worked example.
- **Narrative and framework transparency (Reviewer 4):** Reorganized the Literature Review around an ECR-first narrative detailing single-agent failure modes; added Section 3.2 (Inter-Agent Communication Protocol) and a structural justification of the sub-sequence rule; and formalized the state parameters and agent operational boundaries.
- **Formal complexity and efficiency analysis (Reviewer 4):** Added Section 3.4 deriving $T_{\mathrm{orch}} = O(K \cdot |\mathcal{A}|)$, and expanded Section 4.4 with a token-overhead study (Table 3, Figure 5) reporting LLM calls and token consumption per paradigm.
- **Experimental interpretation and conclusion (Reviewer 4):** Clarified the solver-execution variation across equivalent encodings (Section 5.4); decomposed the 12.18% cost reduction to isolate the lateral-land-route effect (Section 5.3); repositioned human-in-the-loop as a core architectural layer (Section 6); and incorporated four reviewer-suggested references.

A detailed point-by-point response is provided in the accompanying Response Letter. We believe the revised manuscript is now suitable for publication in *Applied Soft Computing*.

Sincerely,

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